

EXACT BASINS AND FIBRES FOR PARALLEL LEAF PEELING ON LABELLED ROOTED FORESTS

ANONYMOUS

ABSTRACT. On every rooted forest carried by a subset of $[n]$, delete all nonroot leaves in parallel and keep the roots. We determine the resulting finite dynamical system. Its depth is the forest height, its 2^n recurrent states are fixed, and, for $n \geq 2$, its sharp maximum depth is $n - 1$, attained by exactly $n!$ states. (For $n = 0, 1$ the maximum depth is zero.) A fixed root set of size r has basin

$$\sum_{k=0}^{n-r} \binom{n-r}{k} r(r+k)^{k-1}.$$

Nested exponential generating functions give every bounded-depth part of every basin. We also determine every local indegree: if a target has m vertices and s nonroot leaves, its one-step fibre has size

$$\sum_{j=0}^s (-1)^j \binom{s}{j} (m-j+1)^{n-m}.$$

A peeling clock, determinant/species basin controls, and a separate target-leaf inclusion–exclusion together close the global basins, all transient shells, local fibres, periodic census, and parameter recovery. Parallel leaf removal, Cayley-forest counts, height enumeration, and the general tools are inputs; the residual internal scope is only this finite-map conjunction.

1. THE SYSTEM AND THE COMPLETE STATEMENT

Fix $n \geq 0$ and write $[n] = \{1, \dots, n\}$. A state is a pair $F = (S, p)$ with $S \subseteq [n]$ and $p : S \rightarrow S$ such that every orbit of p reaches a fixed point. Equivalently, p orients a rooted forest toward its roots

$$R(F) = \{v \in S : p(v) = v\}.$$

A child of v is a $w \neq v$ with $p(w) = v$. Let $L(F)$ be the nonroot vertices having no children. Our update is

$$(1.1) \quad T(F) = (S \setminus L(F), p|_{S \setminus L(F)}).$$

The restriction is well defined: a deleted leaf points to a vertex that is not deleted. The empty forest is allowed. Define the forest height $H(F)$ to be the maximum root–vertex distance when F is nonempty and set $H(\emptyset) = 0$; thus an edgeless forest, including the empty one, has height zero.

For a fixed $R \subseteq [n]$, let E_R denote the edgeless forest on R . For $r \geq 1$ define

$$(1.2) \quad B_{n,r} = \sum_{k=0}^{n-r} \binom{n-r}{k} r(r+k)^{k-1},$$

$$(1.3) \quad A_0(x) = 1, \quad A_h(x) = \exp(x A_{h-1}(x)),$$

$$(1.4) \quad B_{n,r}^{(h)} = \sum_{k=0}^{n-r} \binom{n-r}{k} k! [x^k] A_h(x)^r.$$

The $k = 0$ term in (1.2) is one. Put $B_{n,0} = B_{n,0}^{(h)} = 1$ for $h \geq 0$, and put $B_{n,r}^{(-1)} = 0$ for every $0 \leq r \leq n$.

Theorem 1 (Complete temporal census). *For every $n \geq 0$ the following statements hold.*

- (i) *Every orbit terminates at $E_{R(F)}$, and its entry time is the maximum root–vertex distance in F , namely $H(F)$ under the empty-state convention above.*
- (ii) *The basin of each E_R , $|R| = r$, has size $B_{n,r}$; exactly $B_{n,r}^{(h)}$ of its states have entry time at most h . Hence its exact depth- h shell is $B_{n,r}^{(h)} - B_{n,r}^{(h-1)}$ for every $h \geq 0$.*
- (iii) *If a target G has m present vertices and s nonroot leaves, then*

$$(1.5) \quad |T^{-1}(G)| = \sum_{j=0}^s (-1)^j \binom{s}{j} (m-j+1)^{n-m}.$$

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This includes the empty target, whose only preimage is itself.

(iv) The phase size, periodic data, and Artin–Mazur zeta function are

$$(1.6) \quad |\mathcal{F}_n| = \sum_{m=0}^n \binom{n}{m} (m+1)^{m-1}, \quad |\text{Fix}(T^q)| = 2^n \ (q \geq 1), \quad \zeta_T(z) = (1-z)^{-2^n}.$$

For $n \geq 2$ the largest entry time is $n-1$, and precisely $n!$ states have this depth.

Consequently any one of the fixed-count sequence, the zeta exponent, or the functional graph determines n .

The constituent tools and the temporal primitive are classical. Cayley’s tree count and its rooted-forest extensions are standard [4, 9], with the all-minors determinant supplied by [5]; labelled-tree height enumeration predates the species formulation [11, 10], while nested height generating functions fit the general labelled-species framework [3, 6]. Leaf erasure as a height-governed pruning dynamics is explicit in [7]; the literal parallel *RAKE* primitive appears in [8]; and recent parallel leaf stripping on random recursive trees appears in [1]. We use the usual dynamical zeta convention [2]. We assign no novelty or priority credit to the height clock, semigroup/absorption mechanism, Cayley counts, height EGFs, inclusion–exclusion, zeta conversion, or elementary Hamilton-path extremal argument. The residual internal result is only the simultaneous finite-map package in [theorem 1](#), especially the endpoint-resolved depth series and local-fibre statistic (1.5).

2. PEELING TIME AND THE RECURRENT CORE

For a vertex v , let $H_F(v)$ be the maximum length of a directed path that starts at a descendant leaf and ends at v ; take $H_F(v) = 0$ when v has no nontrivial descendants. This is a subtree height, not the distance of v from its root.

Lemma 2 (Peeling clock). *A nonroot vertex v is deleted in round $H_F(v) + 1$. Therefore $T^h F$ is fixed if and only if $H(F) \leq h$.*

Proof. Induct on $H_F(v)$. A nonroot vertex of height zero is a leaf and is deleted in round one. If $H_F(v) = a > 0$, at least one child has height $a-1$, every child has height at most $a-1$, and induction says that the last child disappears in round a . Thus v becomes a leaf precisely for round $a+1$. The final nonroot deletion in a nontrivial component occurs beside its root on a longest root–leaf path, in the round equal to that path length. An edgeless or empty forest is already fixed and has $H(F) = 0$ by convention. $\square$

Roots never move, while a nonfixed state loses a vertex. Hence there are no nontrivial cycles and every endpoint is $E_{R(F)}$. Conversely every E_R is fixed, giving 2^n fixed states and all periodic assertions in (1.6). The zeta identity follows from $\exp(\sum_{q \geq 1} 2^n z^q / q) = (1-z)^{-2^n}$.

For $n \geq 2$, any forest on at most n vertices has height at most $n-1$. Equality forces one component to contain all n labels along one root–leaf path, so the state is a rooted Hamilton path whose root is an endpoint. Listing its vertices from the opposite leaf to the root is a bijection with permutations of $[n]$. This proves the sharp depth and the $n!$ deepest count for $n \geq 2$. For $n = 0$ the unique empty state has depth zero; for $n = 1$ both the empty state and the singleton-root state have depth zero.

3. GLOBAL BASINS BY TWO ENUMERATIVE ROUTES

Fix $R \subseteq [n]$, $|R| = r > 0$, and first choose a set of k nonroot labels. For $k = 0$ there is exactly the edgeless forest E_R . For $k \geq 1$, the all-minors matrix-tree theorem applied to the complete graph on $m = r + k$ vertices counts forests oriented to the specified roots [5]. Standard all-minors statements orient away from roots; reversing every edge of the undirected complete graph bijects those forests with our parent maps oriented toward roots. The count is

$$(3.1) \quad \det(mI_k - J_k) = r m^{k-1}.$$

Indeed $mI_k - J_k$ has eigenvalue $r = m - k$ on the all-ones line and eigenvalue m on its $(k-1)$ -dimensional complement. Formula (1.2) follows after choosing the k labels. For arbitrary roots on a fixed m -set, summing (3.1) gives $(m+1)^{m-1}$, and then choosing the m present labels gives the phase count in (1.6); its $m = 0$ term is the empty state and equals one. Thus the basin partition also yields the exact identity

$$(3.2) \quad \sum_{r=0}^n \binom{n}{r} B_{n,r} = \sum_{m=0}^n \binom{n}{m} (m+1)^{m-1}.$$

There is a second route that retains time. Give one root a fixed unlabelled marker and label all other vertices. A height-zero tree is just its root. A height-at-most- h tree is a set of child-rooted trees of height at most $h-1$, which gives exactly (1.3). The r distinguished roots support a root-indexed product of r such species. Therefore, on a prescribed set of k nonroot labels, the count is

$$(3.3) \quad k! [x^k] A_h(x)^r.$$

Choosing these labels in $[n] \setminus R$ proves (1.4); lemma 2 converts height into entry time. Coefficientwise, A_h stabilizes to the rooted labelled-tree series as $h \rightarrow \infty$, so (3.3) also recovers (3.1). Thus the determinant and species arguments are independent controls on the complete basin.

4. EVERY LOCAL FIBRE

Let $G = (S, p)$ have $m = |S|$ and let Q be its set of s nonroot leaves. If $m = 0$, then G is empty and its only predecessor is empty: every root or nonroot parent-map vertex would survive one update. Hence assume $m \geq 1$. Suppose $T(F) = G$ and put $L = V(F) \setminus S$. Every element of L must be a leaf of F , so its parent lies in S : if one new vertex pointed to another, the latter would have a child and would survive. Conversely, every nonroot leaf of G must receive at least one new child, or it too would be deleted. No other condition is needed, since roots are immortal and every other nonroot vertex of G already has a child in G .

For a specified ℓ -set $L \subseteq [n] \setminus S$, inclusion–exclusion therefore gives

$$(4.1) \quad \#\{L \rightarrow S : \text{every element of } Q \text{ is hit}\} = \sum_{j=0}^s (-1)^j \binom{s}{j} (m-j)^\ell.$$

Summing over the choice of L and using the binomial theorem yields

$$\sum_{j=0}^s (-1)^j \binom{s}{j} \sum_{\ell=0}^{n-m} \binom{n-m}{\ell} (m-j)^\ell = \sum_{j=0}^s (-1)^j \binom{s}{j} (m-j+1)^{n-m},$$

which is (1.5). This also proves that the local indegree depends on the target only through (m, s) , despite the nonuniform global basins.

5. SCOPE, COLLISION FIREWALL, AND EXACT CONTROLS

The closest internal deletion motif is cycle-minimum pruning on permutations. That system keeps permutation cycles as its components and deletes one distinguished element per nontrivial cycle; its clock is a cycle-length statistic and its counts use permutation assembly. Here the phase is a union of labelled rooted-forest spaces, the update deletes all exposed nonroot leaves, the clock is height, and the counts use Cayley determinants, height species, and a target-leaf inclusion–exclusion. Thus the systems and proof engines are different even though both are finite absorbers.

A bounded owner audit also located direct external sources for parallel *RAKE* [8], height-driven dynamical pruning [7], recursive parallel leaf stripping [1], labelled height enumeration [11, 10], and the all-minors determinant [5]. These mechanisms, enumerations, absorption observations, fixed/zeta conversion, and Hamilton-path extremality are explicit inputs, not claimed advances. The residual description is only the endpoint-indexed assembly and elementary (m, s) fibre calculation. The bounded search did not establish priority for either, and no absence-of-owner inference is made.

The accompanying deterministic verifier enumerates every parent function whose only cycles are loops on all subsets of $[n]$ for $0 \leq n \leq 6$. It compares literal updates and functional-graph fibres with (1.2)–(1.6), expands the nested EGFs with exact rational arithmetic, checks every endpoint/depth cell, and identifies every deepest state. The canonical run contains 400,105 exact assertions; the $n = 6$ lane alone contains 26,830 states. These computations fix conventions and attack boundary cases; the proofs above carry the arbitrary- n statements.

External posting, priority, and submission remain on hold.

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